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BATCH-11

LAB-4 (OOPS)

POLYMORPHISM:

Code 1:

```
class subject {
    int sum(int a, int b) {
        return a+b;
    }

    int sum(int c, int d, int e) {
        return c+d+e;
    }
}

class Student extends subject {
    @Override
    int sum(int a, int b) {
        return a-b;
    }

    int display() {
        int v = super.sum(4, 5);
        return v;
    }
}

public class Polymorphism {
    public static void main(String[] args) {
        subject d = new subject();
        Student e = new Student();
        System.out.println(e.sum(2, 3));
        System.out.println(e.display());
    }
}
```

Output:

```
● PS C:\Users\HP\Desktop\OOPS_590016337> cd "Lab 4"
● PS C:\Users\HP\Desktop\OOPS_590016337\Lab 4> javac Polymorphism.java
● PS C:\Users\HP\Desktop\OOPS_590016337\Lab 4> java Polymorphism
-1
9
```

Code 2:

```
class Animal {

    // Method Overloading
    void sound() {
        System.out.println("Animal makes a sound");
    }

    void sound(String type) {
        System.out.println("Animal sound type: " + type);
    }
}

// Method Overriding
class Dog extends Animal {

    @Override
    void sound() {
        System.out.println("Dog barks");
    }
}

public class Polymorphism1 {

    public static void main(String[] args) {

        // Runtime Polymorphism
        Animal a = new Dog();
        a.sound();

        // Compile-time Polymorphism
        a.sound("Loud");
    }
}
```

Output:

```
● PS C:\Users\HP\Desktop\OOPS_590016337\Lab 4> javac Polymorphism1.java
● PS C:\Users\HP\Desktop\OOPS_590016337\Lab 4> java Polymorphism1
  Dog barks
  Animal sound type: Loud
○ PS C:\Users\HP\Desktop\OOPS_590016337\Lab 4> █
```